

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Led hazard analysis and risk mitigation strategies for enterprise **Java Spring Boot** and **.NET** applications in production, identifying and documenting patient safety hazards through cross-functional collaboration with operations and quality teams; ensured all design changes maintained alignment with **ISO 13485**, **ISO 14971**, and **IEC 62304** regulatory standards.
- Conducted comprehensive risk analyses for software design and process changes, including severity and occurrence assessments, risk–benefit evaluations, and failure mode analysis; developed and maintained risk management documentation with full traceability of design verification and validation activities across **3.5 years** of production support for regulated enterprise systems.
- Designed and executed reliability and safety engineering studies using **Design of Experiments (DOE)**, **Design for Reliability (DfR)**, and statistical process control techniques to evaluate system performance and enhance product robustness; implemented corrective and preventive actions (CAPA) aligned with safety and reliability standards.
- Supported design, maintenance, and continuation engineering of released software systems through production incident investigation and root-cause analysis; independently identified system failures, hazards, and harming scenarios using **Failure Tree Analysis (FTA)**, **Hazard Analysis and Loss (HAL)**, and **Design Failure Mode and Effects Analysis (DFMEA)** methodologies.
- Developed automated reliability monitoring dashboards and **Power BI** reports to track safety metrics, design verification status, and regulatory compliance across interconnected systems; integrated data from **Python** ETL pipelines, **SQL** databases, and **AWS CloudWatch** to provide data-driven risk–benefit insights to stakeholders.
- Applied **Agile** methodologies in cross-functional teams to design and implement reusable safety patterns and innovation frameworks; leveraged **Java**, **Python**, **JavaScript**, **Angular**, and **REST APIs** to develop scalable solutions that increased team velocity and reduced safety rework by **20%**.

SKILLS

Safety & Reliability Engineering – ISO 14971 Risk Management, ISO 13485 Medical Device Quality, IEC 62304 Software Development, Hazard Analysis and Loss (HAL), Failure Tree Analysis (FTA), Design Failure Mode and Effects Analysis (DFMEA), Design of Experiments (DOE), Design for Reliability (DfR), Design for Six Sigma (DFSS), robust design principles.

Software Development & Architecture – Java Spring Boot, C++, C#, .NET, Python, JavaScript, HTML, Angular, REST API design, microservices architecture, CI/CD pipelines (Jenkins, Git), design verification and validation.

Regulated Industry & Medical Device Knowledge – Implantable medical device workflows, design control processes, traceability matrices, change-control management, risk management files, regulatory compliance documentation, post-market surveillance.

Data Analysis & Reporting – Power BI dashboards, Tableau visualizations, SQL (PostgreSQL, MySQL, Oracle), Python (pandas, NumPy, scikit-learn), statistical analysis, Design for Manufacturability (DRM), Design for Lean Sigma (DFLS), process control metrics.

Cloud & DevOps Tools – AWS (EC2, S3, Lambda, CloudWatch), Docker containerization, Linux, shell scripting, GitHub Actions, Vercel deployment, monitoring and alerting infrastructure.

Project Management & Collaboration – Agile methodologies, Jira ticketing, Confluence documentation, cross-functional team leadership, stakeholder communication, UAT coordination, design input and output alignment.

Artificial Intelligence & Automation – AI tools (Claude API, Anthropic SDK), prompt engineering, workflow automation, constraint guardrails, structured-JSON design, end-to-end system orchestration.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
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BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021
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PROJECTS

Cardiac Payments Operations Dashboard & Risk Management System (HCL Technologies)

- Designed and maintained a comprehensive safety and risk management dashboard for a regulated cardiac medical device payments platform, integrating **ISO 14971 risk assessments**, **ISO 13485 design controls**, and **IEC 62304 software lifecycle documentation** into a single unified system.
- Conducted hazard analysis and loss (HAL) studies across payment message flows, identifying and mitigating safety-critical failure modes; developed **Failure Tree Analysis (FTA)** and **Design Failure Mode and Effects Analysis (DFMEA)** documentation to support design verification and validation traceability.
- Built production monitoring dashboards using **Power BI**, **Python ETL pipelines**, and **SQL** to track safety metrics, design control status, and regulatory compliance across interconnected enterprise systems; reduced risk management review cycles by **25%** through automated data-driven reporting.

Reliability & Safety Engineering Automation Platform (Personal & HCL Technologies)

- Architected an end-to-end reliability and safety engineering workflow using **Claude API (Anthropic)** and prompt engineering to automatically generate **Design of Experiments (DOE)** study designs, failure analysis reports, and risk mitigation recommendations from design specifications.
- Implemented **GitHub Actions** CI/CD pipelines and **Vercel** deployment to orchestrate automated **Design for Reliability (DfR)** assessment generation, **Design Failure Mode and Effects Analysis (DFMEA)** templates, and safety pattern libraries as reusable artifacts.
- Developed structured-JSON prompt engineering with constraint guardrails to ensure safety documentation met **ISO 14971** and **ISO 13485** compliance requirements; integrated **Java Spring Boot** backend, **Angular** frontend, and **Python** automation layers to reduce manual safety engineering effort by **40%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)